

Curriculum: Wissenschaftliche Methodik 2026

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Studiendesign und Statistik

- Day 1: Clinical Epidemiology & Study Design
- Day 2: Introduction to Biostatistics & Hands-on in R
- Day 3: Introduction to Resampling Statistics & Hands-on in R

- Download course material and Tutorial from https://github.com/statscourse26 (Public Repositories)
- For Day 2 & 3 everybody should have access to wlan and Posit Cloud (online version of R) posit.cloud
- You can use an AI as an R manual



StatsCourse26

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main | 1 Branch | 0 Tags | Go to file | Code

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01-Introduction to Biostatistics.pdf	Rename Introduction to Biostatistics.pdf to 01-Introductio...	5 days ago
02-Descriptive Statistics.pdf	Rename Descriptive Statistics.pdf to 02-Descriptive Statis...	5 days ago
03 - Hypothesis Testing	Rename Introduction to Statistics 1 .pdf to 03 - Hypothesi...	yesterday
04 - Univariate Tests	Rename Introduction to Statistics 2.pdf to 04 - Univariate ...	yesterday
05 - Kaplan Meier Survival Analysis.pdf	Rename kaplanmeier.pdf to 05 - Kaplan Meier Survival An...	3 days ago
06 - Assessing Normality.pdf	Rename Assessing Normality.pdf to 06 - Assessing Norma...	yesterday
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Tutorial - Resampling Statistics.pdf	Rename Resampling Statistics.pdf to Tutorial - Resampling...	3 days ago
day2-exercises.R	Add files via upload	yesterday
day3-exercises.R	Add files via upload	3 days ago

About

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Languages

R 100%

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- New Jupyter Project
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File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function | Addins | R 4.6.1

day2-exercises.R

Run | Source

```
1  # ============================================================================
2  #  Wissenschaftliche Methodik 2026 - Day 2
3  #  Thirty minutes of hands-on
4  #
5  #  No data files and no packages: the numbers are the ones from the
6  #  lecture. Run it line by line (Ctrl/Cmd + Enter) and read the output
7  #  before moving on. Four tasks, marked ### YOUR TURN, with the answer
8  #  just below each.
9  # ============================================================================
10
11 # The twenty patients from the lecture: chronic periodontitis, ten per
12 # protocol, reduction in probing depth (mm) after six months. Fixed here
13 # so that everything you compute matches the slides.
14 a <- c(2.1, 2.9, 1.8, 3.0, 2.3, 1.4, 0.8, 3.4, 2.6, 1.9)   # protocol A
15 b <- c(1.6, 0.8, 2.1, 2.8, 1.7, 0.9, 0.1, 2.2, 1.5, 3.5)   # protocol B
16 a; b
17
```

144:1 (Untitled) R Script

Console | Terminal | Background Jobs

R 4.6.1 · /cloud/project/

```
>
```

Environment | History | Connections | Tutorial

Import Dataset | 293 MiB | List

R | Global Environment

Environment is empty

Files | Plots | Packages | Help | Viewer | Presentation

Folder | File | Upload | Delete | Rename

Cloud > project

Name	Size	Modified
..		
.Rhistory	0 B	Aug 2, 2026, 12:45 PM
day2-exercises.R	6.6 KB	Aug 7, 2026, 12:41 AM
day3-exercises.R	5.1 KB	Aug 6, 2026, 1:58 AM
project.Rproj	205 B	Aug 8, 2026, 8:18 PM
Stats.R	0 B	Aug 6, 2026, 1:40 AM



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The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, **Windows and Mac** users most likely want one of these versions of R:

- Download R for Linux (Debian, Fedora/Redhat, Ubuntu)
- Download R for macOS
- Download R for Windows

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2026-06-24, Happy Hop) R-4.6.1.tar.gz, read what's new in the latest version.
- The CRAN directory src/base-prerelease contains R alpha, beta, and rc releases as daily snapshots in time periods before a planned release.
- Between releases, the same directory src/base-prerelease contains snapshots of current patched and development versions. Please read about new features and bug fixes before filing corresponding feature requests or bug reports.
- Alternatively, daily snapshots are available here.
- Source code of older versions of R is available here.
- Contributed extension packages.

Questions About R

- If you have questions about R like how to download and install the software, or what the license terms are, please read our answers to frequently asked questions before you send an email.

Supporting CRAN

- CRAN operations, most importantly hosting, checking, distributing, and archiving of R add-on packages for various platforms, crucially rely on technical, emotional, and financial support by the R community.

https://cran.r-project.org/index.html

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RStudio IDE

Direct download links for RStudio Desktop 2026.07.1, the version this documentation describes:

Platform	Download
Windows	RStudio-2026.07.1-147.exe
Windows (zip archive)	RStudio-2026.07.1-147.zip
macOS 13+	RStudio-2026.07.1-147.dmg
Ubuntu 22 / 24, Debian 12 / 13	rstudio-2026.07.1-147-amd64.deb
Ubuntu / Debian (tar.gz)	rstudio-2026.07.1-147-amd64-debian.tar.gz
RHEL 9 / 10, Fedora	rstudio-2026.07.1-147-x86_64.rpm
RHEL / Fedora (tar.gz)	rstudio-2026.07.1-147-x86_64-fedora.tar.gz

RStudio Server

Direct download links for RStudio Server 2026.07.1, the version this documentation describes:

Platform	Download
Ubuntu 22 / 24, Debian 12 / 13	rstudio-server-2026.07.1-147-amd64.deb

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https://docs.posit.co/ide/user/#rstudio-ide-oss-downloads

Statistical software in biostatistics

	Strengths	Limitations
R	Free; ~22,000 packages on CRAN; everything is an object; the script is the record	Steep start, no menus, package quality varies
Stata	Menus and commands side by side; excellent manuals; strong in epidemiology and survival	Licence cost; new methods arrive later
Python	NumPy, pandas, SciPy, statsmodels, lifelines; unbeatable for machine learning and automation	Classical biostatistics thinner than in R
SAS	Still the standard for regulatory submissions in pharma; validated	Expensive; losing ground in academia
SPSS	Menu-driven, quick to teach, familiar in clinics	Point-and-click is hard to reproduce

AI-assisted statistics: what it can and cannot do

What it does well

- Writes the code, so your time goes into design and interpretation
- Explains error messages and unfamiliar output
- Lowers the barrier to R for clinicians who would otherwise click through menus
- Posit Assistant reads your live session, so its code fits your actual data

Where it fails

- It does not know your design, your sampling or your confounders
- Confident errors: invented functions, wrong defaults, silently dropped rows -> statistical hallucination
- It will run any analysis you ask for, including the one the data do not support
- Code written for another package version stops working

Use it to write the code, never to choose the method - and read every line before you run it

Posit Assistant sits in RStudio and Positron and runs on Claude; from R itself the ellmer package calls it with chat_anthropic(). The author of the paper is still you.

Key concepts in clinical epidemiology

- Epidemiology
  - To study the occurrence of disease using appropriate methods / using quantification of data
  - Quantitative medicine / clinical epidemiology provides a methodological toolbox

- Research question in clinical medicine / dentistry
  - how often does it occur? Frequency
  - who has the disease? Diagnosis
  - why does it occur? Etiology
  - what will happen in future? Prognosis
  - what can you do about it? Therapy

Research question formulation = Occurrence formulation

- f(Determinants) = Disease
- Looks simple, but if you skip this step, you get stuck later on
- Eg. f(Smoking) = Implant failure
- Eg. f(BP) = Stroke

Quantification of disease

Measures of frequency

- Prevalence
- Cumulative Incidence (Probability)
- Incidence Rate

Measure of association

- Relative Risk (Risk Ratio, Rate Ratio)
- Risk Difference, Rate Difference
- Attributable Risk

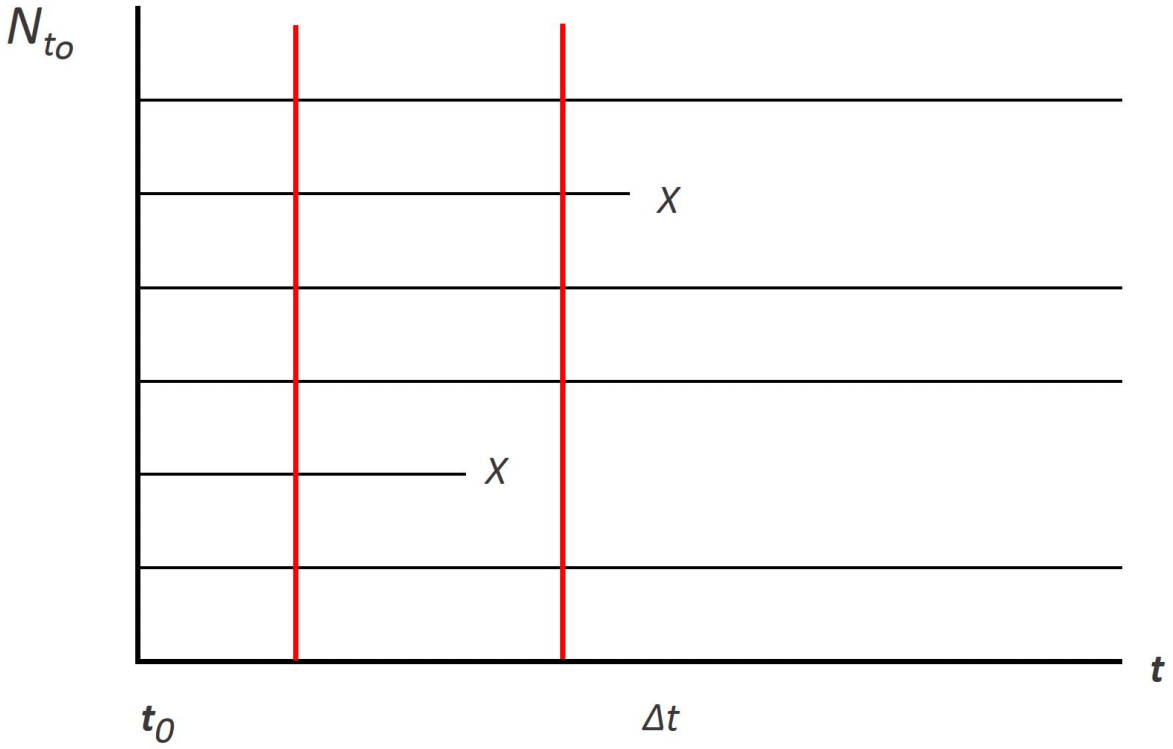
Measure of disease frequency: Prevalence

Prevalence = Existing Cases / Total Population

Proportion / Fraction / Percentage

Ranges from 0 to 1 (or 0% to 100%)

Measure of disease frequencey: Incidence



Cumulative Incidence (CI)

$CI_{\Delta t}$ = New Cases / $N_{t=0}$

Risk

Proportion / Fraction / Percentage
Probability
Ranges from 0 to 1 (or 0% to 100%)

Time specification
Define population at risk
Complete follow-up
Closed cohort

Cumulative Incidence in Medicine

- Absolute Risk
- Case fatality (lethality)
- 5-year Survival
- Attack „rate“
- Lifetime Risk

What is the lifetime risk of death?

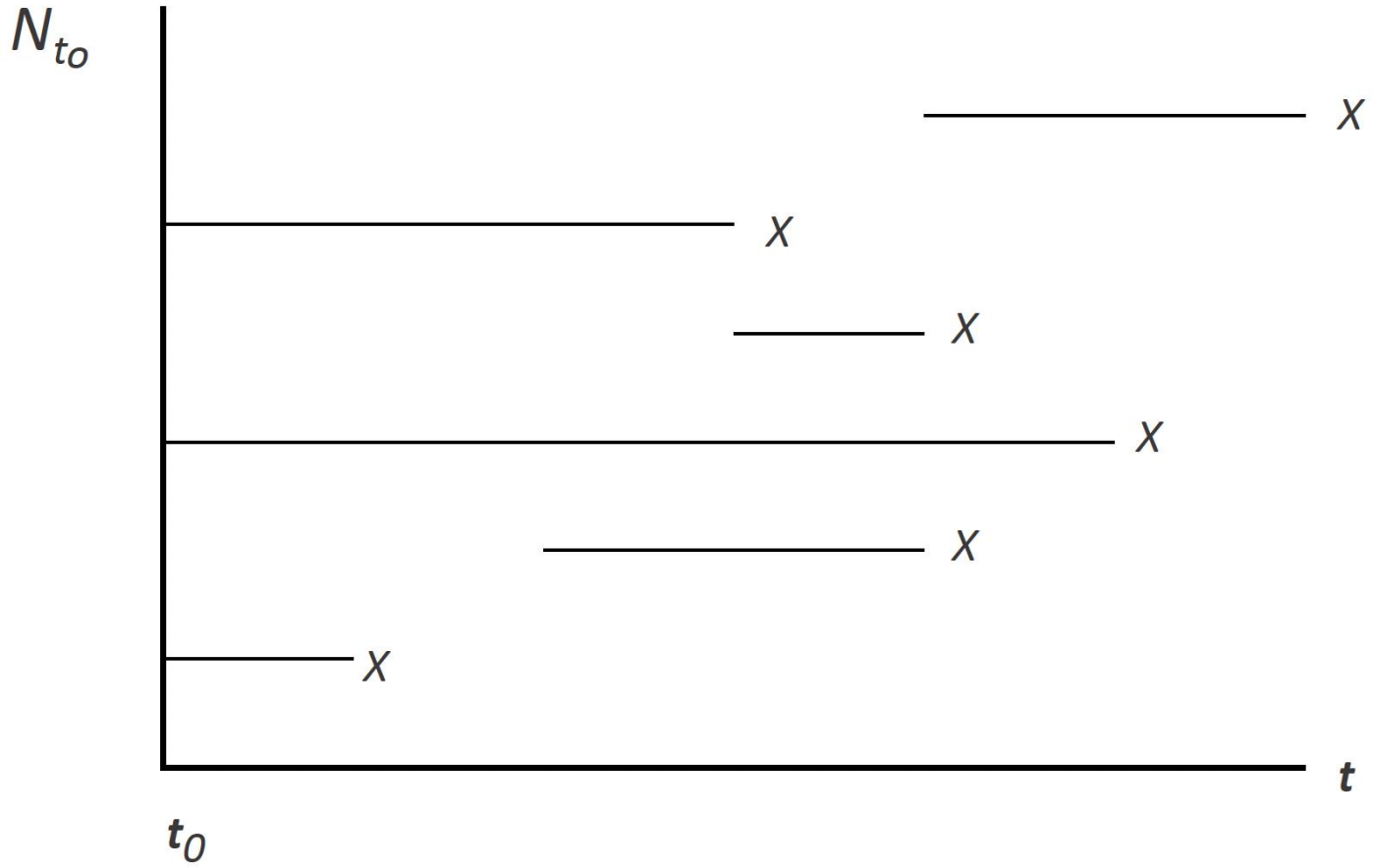
A 0%

B ca. 82 years

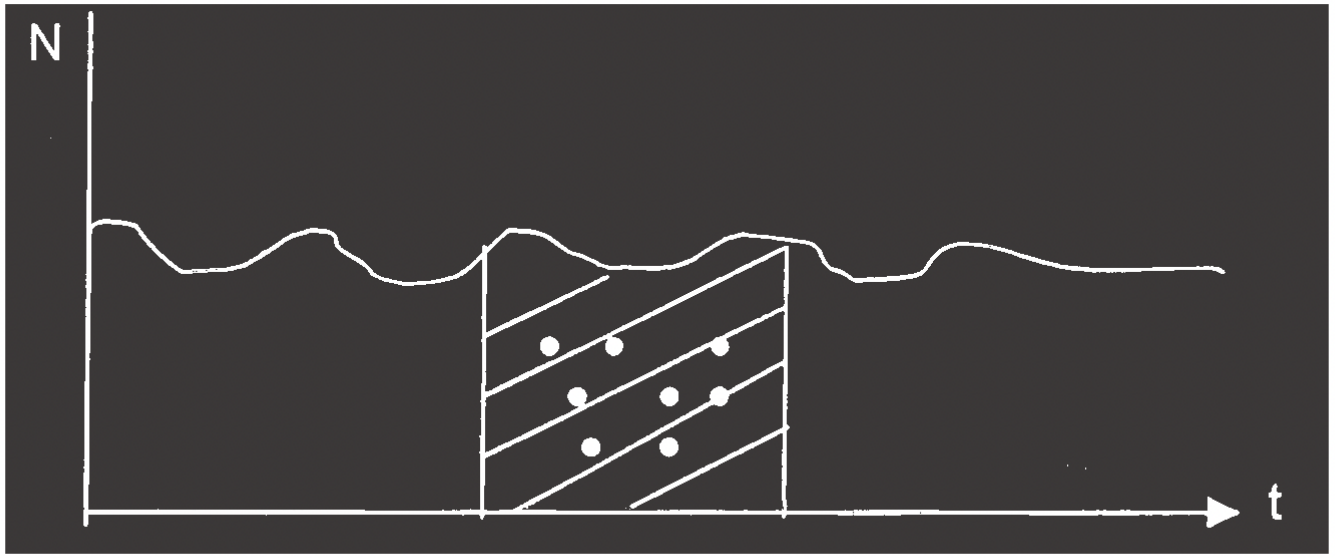
C 100%

D Cannot be calculated

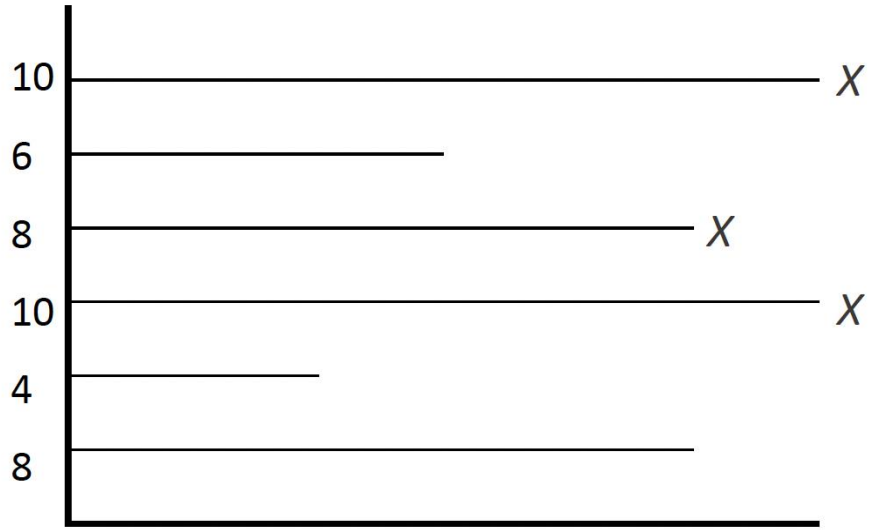
Measure of disease frequencey: Incidence rate (dynamic cohort, observation time may be censored)



Measure of disease frequencey: Incidence rate

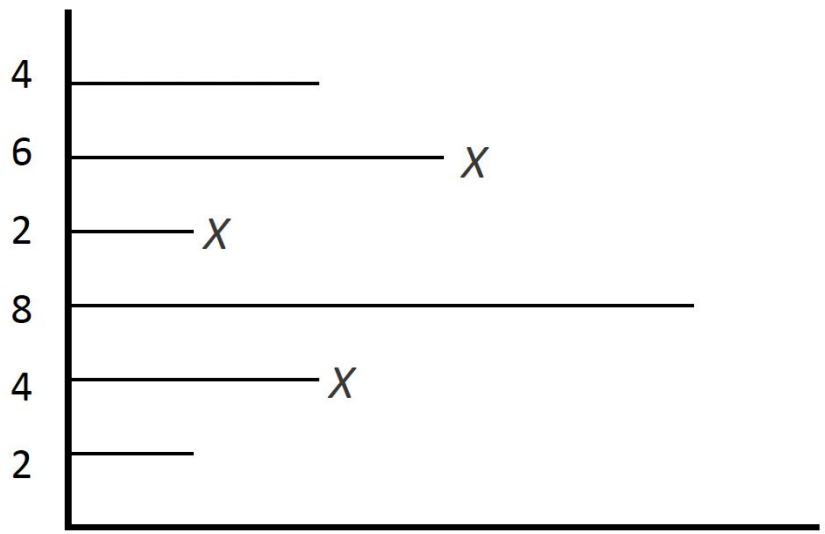


$$Incidence\ rate = {}^{New\ events}/_{Population\text{–}time}$$



3/46 = 0.065 cases per person-year

= 6.5 cases per 100 person-years



3/26 = 0.115 cases per person-year

= 11.5 cases per 100 person-years

Incidence rate

- Density, not a proportion
- Unit in person-years
- Ranges from 0 to >1 (eg 20 cases in 10 patient-years)
- 20 cases in 100 patients, each observed for 0.1 years (-> 10 person-years)
- Life-expectancy = 1 / Incidence rate
- Incidence rate of death 0.011 -> 1/0.011 = 91 years

Relation between CI and IR

$CI_t = 1 - e^{-IR * dt}$

Rate acts on a shrinking pool, exponential decay

IR can be > 1 (multiple events per person year), but CI cannot be > 1

1 - & exponent: ensures CI stays within 0 and 1 (as the negative exponential will never be >1)

Euler number: natural logarithm, use it to model continuous and exponential decay (pool of people that can get a disease shrinks over time)

Relation between CI and IR

Hypothetical example: incidence rate of implant failure in edentulous patients is 0.3 per 100 patient-years from a large observational study

We want to know the 5-year survival probability

$1-e^{-0.003*5} = 1-2.718^{-0.003*5} = 0.015$

*R code: 1-exp(-0.003*5)*

5-year survival is 98.5%

Relation between Prevalence and IR

Prevalence = IR * duration of disease

IR of implant failure in 70-year-old male patients is 0.3 per 100 patient-years

Life expectancy at age 70 is 14 years

Prevalence is 0.003 * 14 = 0.042

4.2%

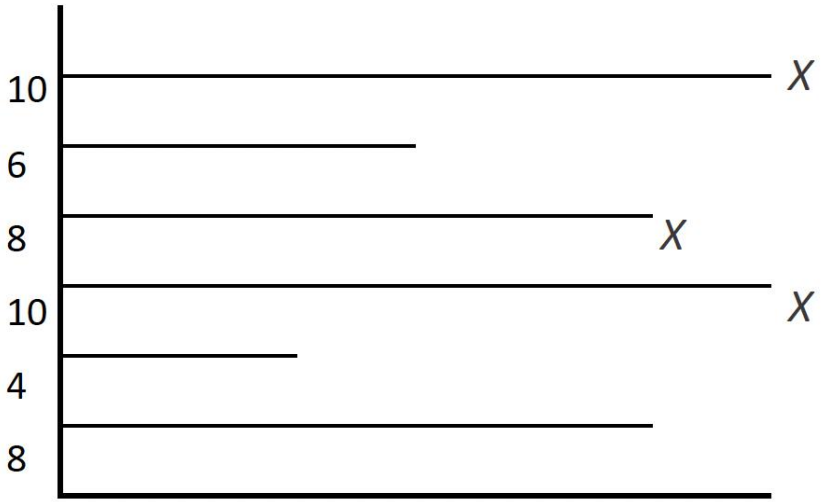
Measures of disease association: Relative Risk (Risk Ratio)

	N	Implant Failures after 10 years	$CI_{10\ years}$
Smoker	239	43	0.180
Non-Smoker	1750	140	0.080
Total	**1989**	**183**	0.092

Hypothetical data from a large scale cohort study

Relative Risk = $Risk_1$ / $Risk_2$ = $CI_{1\ 10\ years}$ / $CI_{2\ 10\ years}$

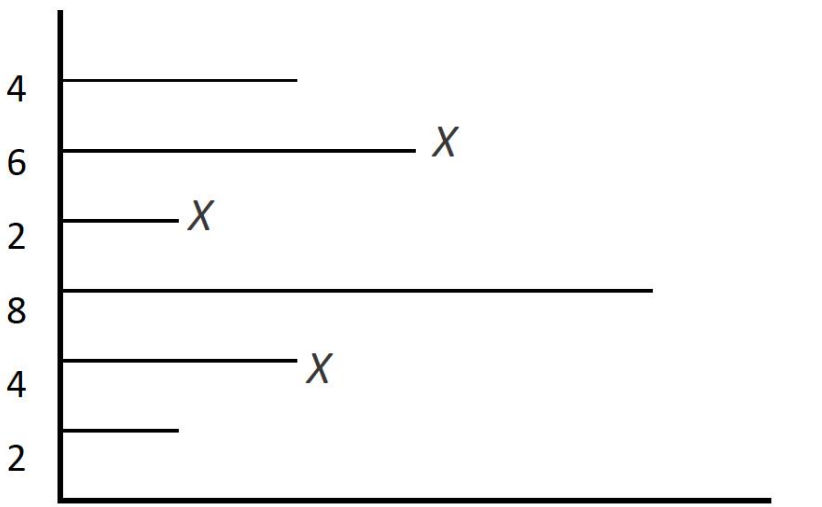
RR = 0.180 / 0.080 = 2.25



$CI_8 = 1/6 = 0.17$

$CI_{10} = 3/6 = 0.5$

IR = 3/46 = 6.5 per 100 py



$CI_8 = 3/6 = 0.5$

$CI_{10} = 3/6 = 0.5$

IR = 3/26 = 11.5 per 100 py

Relative Risk:

$CIR_8 = 0.5 / 0.17 = 2.9$

$CIR_{10} = 0.5 / 0.5 = 1$

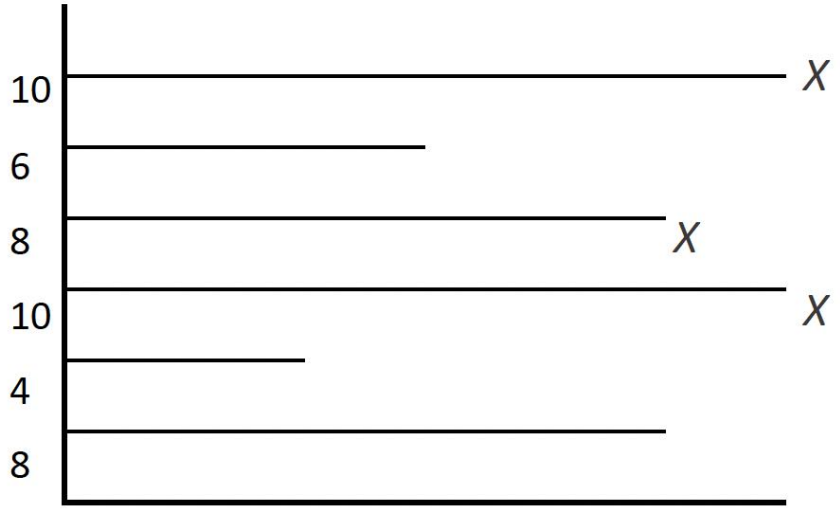
IRR = 11.5 / 6.5 = 1.8

Relative Risk

- Ranges from 0 to infinity
- Null value = 1
- Long follow-up: CIR approaches 1
- Short follow-up: CIR approaches IRR (Incidence Rate Ratio)

Measure of disease association: Risk Difference

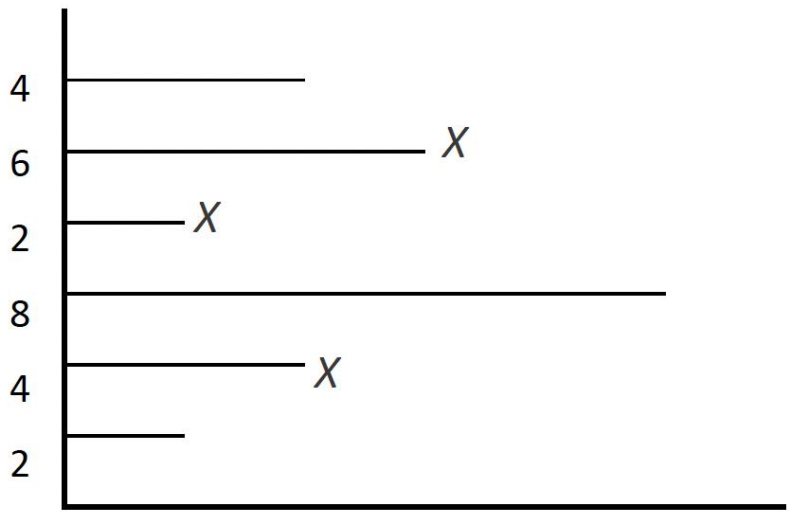
Ranges from -1 to 1
Null value = 0
Special instance: 1 / RD = NNT to prevent one event
Example:
10-year Risk of implant failure in smokers versus non-smokers
$R_1 = 0.179$, $R_2 = 0.080$, RD = 0.099
NNT = 1/0.099 = 10 patients, for every 10 patient that stops smoking we prevent 1 implant failure



$CI_8 = 1/6 = 0.17$

$CI_{10} = 3/6 = 0.5$

IR = 3/46 = 6.5 per 100 py



$CI_8 = 3/6 = 0.5$

$CI_{10} = 3/6 = 0.5$

IR = 3/26 = 11.5 per 100 py

Risk difference:

$CID_8 = 0.5 - 0.17 = 0.33$

$CID_{10} = 0.5 - 0.5 = 0$

IRD = 11.5 - 6.5 = 5

Attributable Risk

	N	Implant Failures after 10 years	$CI_{10\ years}$
Smoker	239	43	0.179
Non-Smoker	1750	140	0.080
Total	**1989**	**183**	0.092

How many smokers would have experienced a failure anyway?

239 * 0.08 = 19 failures due to background risk
How many of the 43 failures are due to smoking? 43-19 = 24, 24 / 43=0.553

Attributable risk among exposed 0.179-0.080 / 0.179 = 0.553
55.3% of implant failures in smokers are due to smoking

Attributable Risk

Attributable risk among exposed

Etiologic fraction

Attributable Risk = $(R_1 - R_2) / R_1$

Attributable Risk = (RR – 1) / RR

AR of various risk factors may exceed 100%

How attributable risk is calculated:
Divide numerator and denominator by R2

$$\frac{R_1 - R_2}{R_1} = \frac{\frac{R_1 - R_2}{R_2}}{\frac{R_1}{R_2}} = \frac{\frac{R_1}{R_2} - 1}{\frac{R_1}{R_2}} = \frac{RR - 1}{RR}.$$

Quiz (True or False)

- The prevalence always ranges from 0 to 1
- The 5-year survival is a cumulative incidence
- Prevalence rate can be defined as the total number of new cases per person-years
- Incidence rate, cumulative incidence, and relative risk are all measures of frequency
- The prevalence can be defined as incidence rate * duration of disease
- Rates are proportions

How to set up a study

Experimental

- Intervention study (Trial)
- Investigator assigns exposure
- Write down your occurrence formulation f(Determinants) = Disease !

Observational

- No intervention
- Exposure assigned by nature or participants
- Cross sectional
- Longitudinal
  - Cohort
  - Case-control

Case reports or case-series

- New cases
- Side Effects
- New Diseases



Cross-sectional study

Simultaneous measurement of exposure and outcome

We look at prevalence

Conceptually longitudinal (e.g. prognostic research, association between smoking and peiodontitis)

- Reserach question may be etiology (causal question or occurrence relation)

Conceptually cross-sectional (e.g. diagnostic research, who has the disease)

- Screening is a special case of a diagnostic study on a population level

Meta-analysis is a cross-sectional study, bc unit of analysis are other studies

Cave: reverse causality (bc there is no time lag)

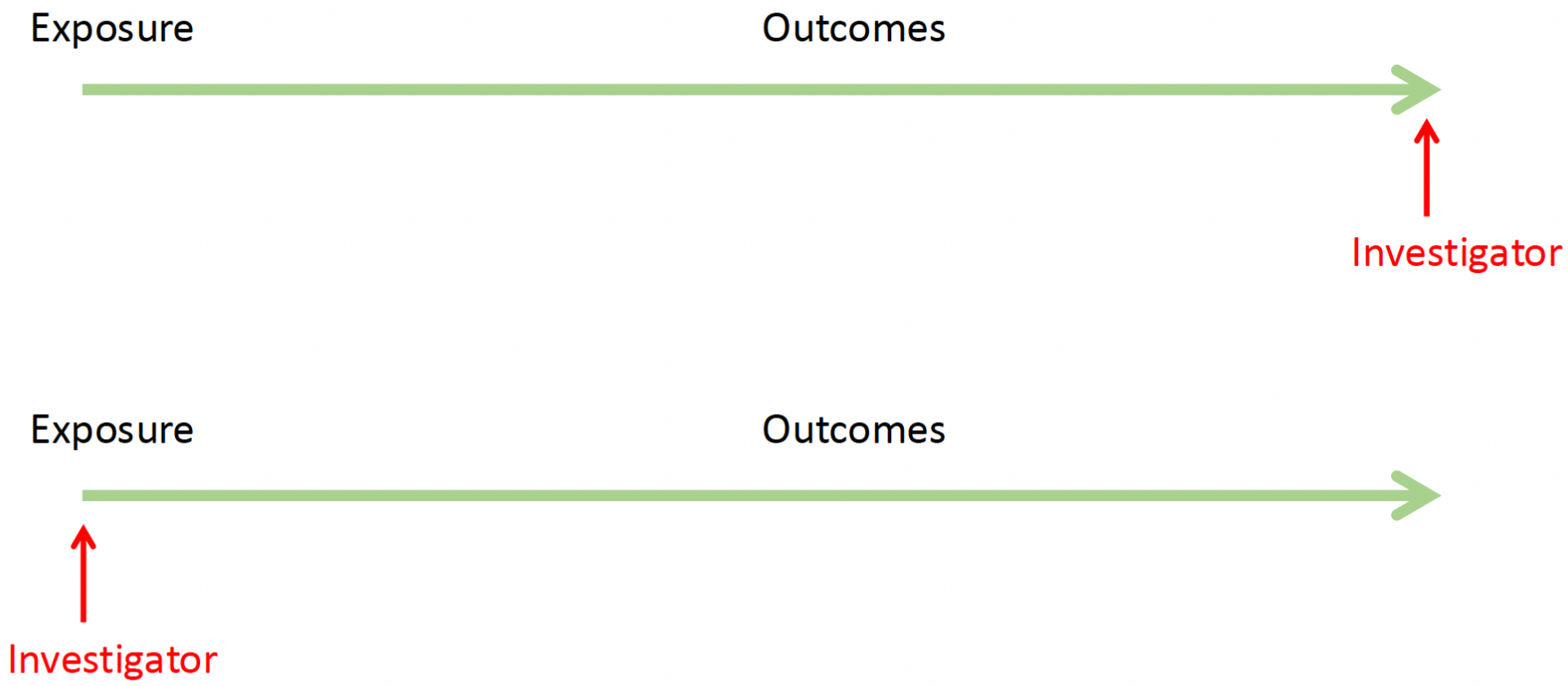
- Direction of cause and effect may be the opposite of what you assume
- E.g. people who visit dentist less often have more tooth loss
- But maybe people who have less teeth find there is no point in visiting the dentist more often

Follow up (cohort) study

- Exposure -> Outcome
- Time aspect
  - Prospective
  - Retrospective
  - Ambispective
- All measures of frequency and association can be obtained

	N	Implant Failures after 10 years	$CI_{10\ years}$
Smoker	239	43	0.180
Non-Smoker	1750	140	0.080
Total	**1989**	**183**	0.092

Cohort study



Cohort study

CLOSED	OPEN
Static population	Dynamic population
Number of persons can only decrease	Date of entry and exits differs
	Number of persons varies
Directly calculate CI and IR	Only calculate IR (CI indirectly via formula)

Cohort study

- start with identifying individuals with and without exposure
- Selection of index (exposed) group

    - Specific exposure
        - Crowns / bridges
        - Implant patient / Edentulous

    - Captive population
        - Physicians / Dentists (e.g. Radiation)
        - Patients in an outpatient clinic

    - General population
        - Population-based / population representative (e.g. water fluoridation)

Strengths and Limitations of a Cohort Study

Strengths

- Rare exposure (bc you start with the exposed versus unexposed group)
- Allows estimation of absolute and relative risk (IR, RR)
- Multiple outcomes from a single exposure (e.g. smoking -> implant failure, periodontitis, tooth loss)
- Temporal relationship clear (exposure before outomes) as opposed to cross-sectional

Limitations

- Not efficient for rare diseases (need large sample size and long follow-up)
- Expensive -> costs of recruitment, follow-up, data collection
- Long time-investment (espceially prospective studies)
- Loss to follow-up -> can introduce attrition bias (sicker patients may be more likely to drop out, measured association may no longer represent the true value)

Large cohort

	N	No Implant failure	Implant failure
Smoker	1000	950	50
Non-Smoker	20‘000	19‘500	500
Total	**21‘000**	**20‘450**	**550**

$$RR = \frac{\text{Risk in Exposed (Smokers)}}{\text{Risk in Unexposed (Non-smokers)}}$$

$$RR = \frac{50/1000}{500/20000} = 2.00$$

Large cohort

	N	No Implant failure	Implant failure
Smoker	a+b	b	a
Non-Smoker	c+d	d	c
Total	**a+b+c+d**	**b+d**	**a+c**

$$RR = \frac{\text{Risk in exposed}}{\text{Risk in unexposed}} = \frac{\frac{a}{a+b}}{\frac{c}{c+d}}$$

From cohort study to case control study

	N	No Implant failure	Implant failure
Smoker	a+b	b	a
Non-Smoker	c+d	d	c
Total	**a+b+c+d**	**b+d**	**a+c**

$$RR = \frac{\text{Risk in exposed}}{\text{Risk in unexposed}} = \frac{\frac{a}{a+b}}{\frac{c}{c+d}}$$

From cohort study to case control study

	N	No Implant failure	Implant failure
Smoker	a+b	b	a
Non-Smoker	c+d	d	c
Total	**a+b+c+d**	**b+d**	**a+c**

Rare disease assumption: $a + b \approx b \quad \text{and} \quad c + d \approx d.$

$$RR = \frac{\frac{a}{a+b}}{\frac{c}{c+d}} \overset{\text{rare}}{\approx} \frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a/b}{c/d} :$$ This is the odds ratio OR

From cohort study to nested case-control study

	N	No Implant failure	Implant failure
Smoker	1000	950 = b	50 = a
Non-Smoker	20‘000	19‘500 = d	500 = c
Total	**21‘000**	**20‘450**	**550**

a/c = can be obtained by analysing the cases

b/d = can be obtained from patients without implant failure

Instead of taking the full sample, a random sample (20% b and 20% d) will be sufficient

From cohort study to nested case-control study

	N	No Implant failure	Implant failure
Smoker	1000	950 = b	50 = a
Non-Smoker	20‘000	19‘500 = d	500 = c
Total	**21‘000**	**20‘450**	**550**

a/c = 50/500 = 0.1

b/d = 950/19‘500 = 0.05

OR **= 0.1 / 0.05 = 2 (so it‘s the same as RR)**

Note: (a/b) / (c/d) = ad / bc = (a/c) / (b/d)

Relation between RR and OR

For rare diseases (prevalence 5%-10%) OR approximates RR

We are NOT interested in OR per se, difficult to interpret

When the disease is NOT rare

- OR will overestimate RR when RR > 1
- OR will underestimate RR when RR < 1

We need background risk in unexposed group to calculate RR from OR:

- OR = odds ratio
- p_0 = risk (probability) in unexposed group

then the risk ratio (RR) is:

$$RR = \frac{OR}{(1 - p_0) + (p_0 \times OR)}$$

Strengths and limitations of case-control studies

Compare the exposure odds among the cases with exposure odds among the controls
Efficient for rare diseases and long latency periods
Quicker and less expensive than cohort studies
Controls can be drawn from a cohort study (nested) or different sources than the cases (e.g. hospital)
Can be prospective or retrospective (but usually retrospective)
Ensure comparability between cases and controls (e.g. matching)
No absolute risks or rates can be obtained
More prone to bias (e.g. selection bias and information bias)
If done properly, difficult to design

Quiz

A randomized controlled trial is per definition a pospective study

A meta-analysis can be observational or experimental or both

In a case control study the risk ratio and the attributable risk can be calcuated

key assumption in a case-control study is that controls are sampled such that the distribution of their exposure reflects that of the population where the cases arise from

All medical research is conceptually longitudinal

Randomised clinical trials

Design	In medicine	In dentistry
Two arms of patients (parallel group)	Statin versus placebo for cardiovascular prevention	Fluoride varnish versus none for caries in children
Two arms of clusters	Hand-hygiene programme randomised by hospital ward	Toothbrushing programme randomised by school
2 x 2 factorial	Aspirin and vitamin E, both tested at once	Chlorhexidine rinse and powered brush, both at once
N = 1 trial	Analgesic versus placebo in alternating blocks, one patient	Desensitising toothpaste versus control in one patient
Cross-over	Two inhalers in stable asthma, with a washout	Two denture adhesives in the same patient, two weeks each
Non-inferiority	A new oral anticoagulant versus warfarin	Short implants versus sinus lift with standard implants

Randomised clinical trial: the design

The investigator assigns the exposure - therefore an experiment

Per definition prospective: you cannot go back in time to randomise

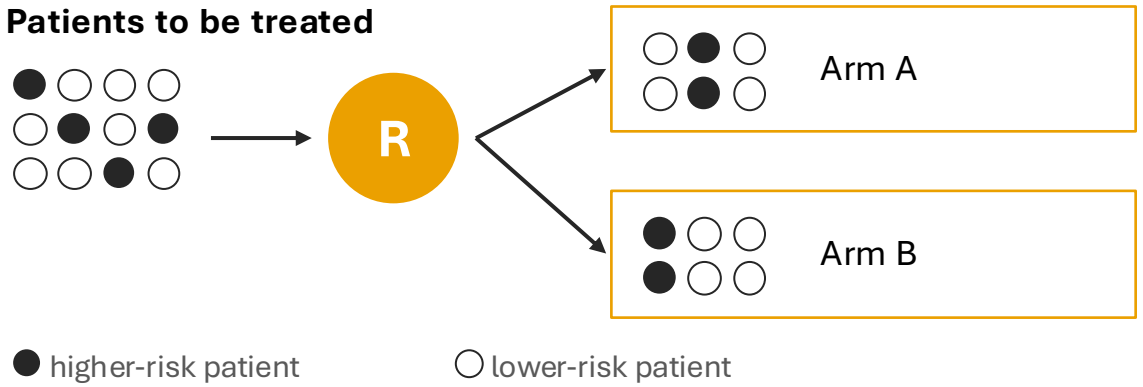
An RCT is a prospective cohort study with randomly allocated exposure

All measures of frequency and association can be obtained

Write the occurrence relation first: f(treatment) = outcome

Interventional study = experimental study = trial. If nature assigns the exposure, the study is observational.

Why randomise?



Both arms end up with the same mix of prognostic factors - the measured ones and, on average, the ones you never measured

Removes confounding by indication - the clinician no longer chooses

Balances unmeasured confounders, on average

Provides the formal basis for the statistical inference

No guarantee: with small samples or a rare confounder the arms can still be unbalanced - always inspect the baseline table.

Equipoise: the ethical prerequisite

The chance that the new treatment is better than the control should be 50%

- Too little prior knowledge: you randomise patients into a question nobody can answer yet
- Too much prior knowledge: withholding the effective treatment is no longer defensible
  - COVID-19 and antimalarial drugs: weak evidence was enough to disturb equipoise
  - A new surgical technique: never randomise the very first patient - start with a case series

Equipoise is not a number you can compute - it is the honest state of the field before the first patient is randomised.

Blinding



Allocation concealment (nobody can foresee the next allocation) protects the randomisation; blinding protects the measurement. Blinding prevents information bias - it does nothing about confounding. Where blinding is impossible, use records or images rather than a questionnaire, scored by an independent assessor.

A few words on p-values

Significance testing rests on the p-value, which mixes effect size with precision

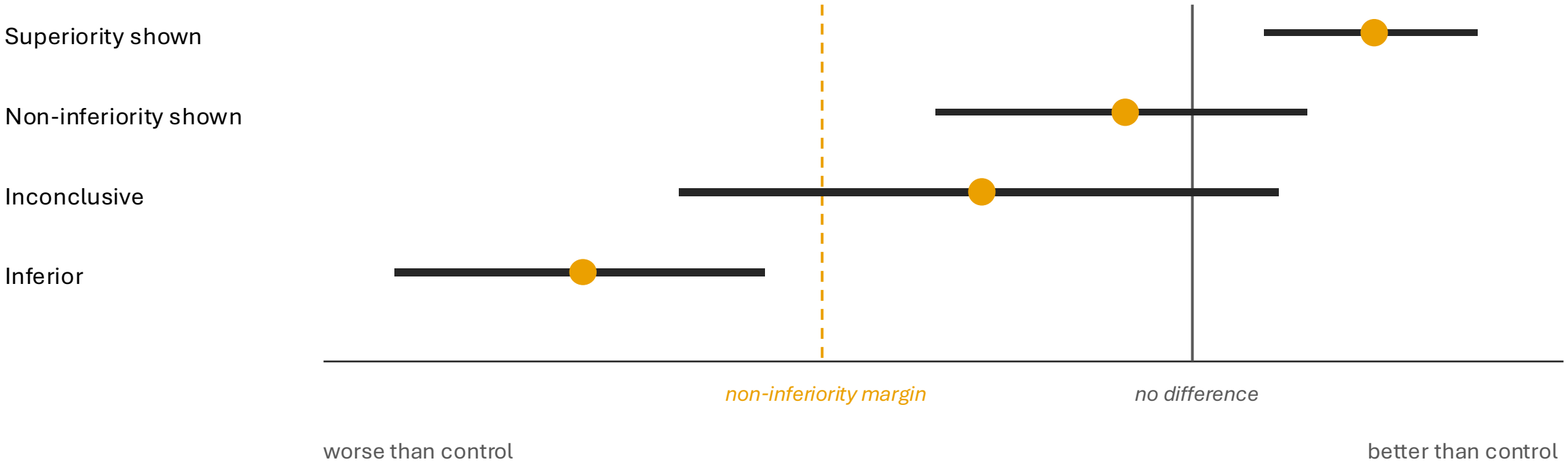
It is not possible to measure two things with one number

It reduces a quantitative result to a qualitative yes / no

Study	Effect estimate	Precision	p-value
A	large	low	0.04
B	small	high	0.04

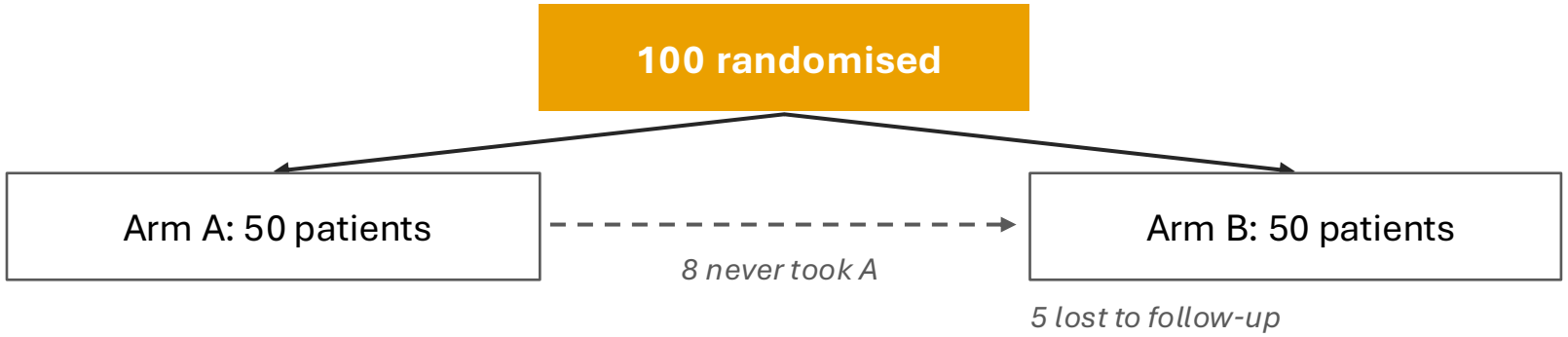
Same p-value, entirely different studies. Present the effect estimate with its confidence interval

Superiority, non-inferiority, inconclusive



The point estimate with its confidence interval answers the question; the p-value does not.

Intention-to-treat or per-protocol?



Intention-to-treat

- 50 versus 50, in the arm they were randomised to
- Randomisation - and comparability - preserved

Per-protocol

- 42 versus 45, only those who completed
- Adherence is prognostic: confounding is back

Strengths and limitations of trials

Strengths

- Randomisation handles unmeasured confounding
- No selection bias at study entry
- Blinding controls information bias
- Exposure is standardised and documented
- Direct answer to a therapeutic question

Limitations

- Expensive, slow, often underpowered
- Selected patients - generalisability suffers
- Rare and late adverse effects are missed
- Many questions cannot be randomised at all
- Ethically bound by equipoise

Quiz: trials (true or false)

A randomised trial is per definition a prospective cohort study

A study is only an experiment if the exposure is allocated at random

Randomisation guarantees that the two arms are comparable

Blinding protects against confounding

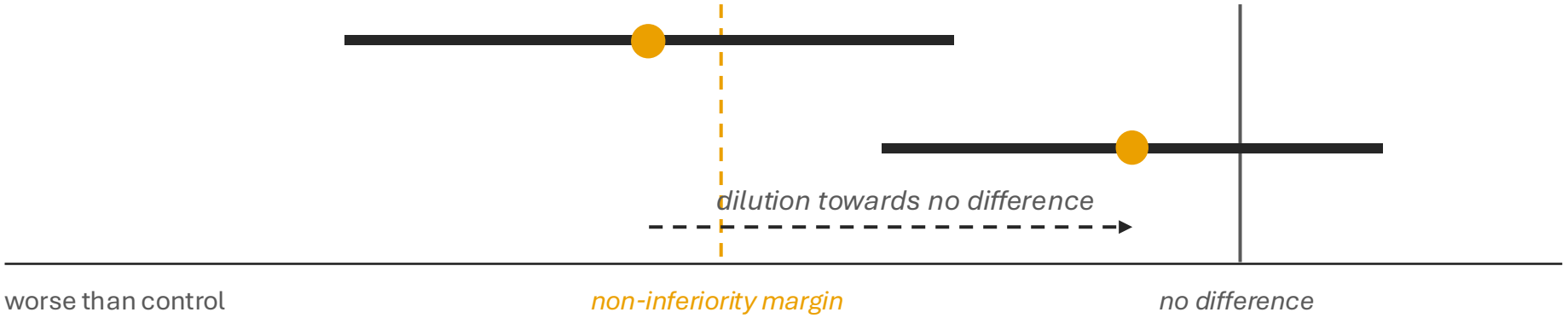
In a trial, selection bias can only arise through loss to follow-up

A per-protocol analysis is the more conservative analysis in a non-inferiority trial

Non-inferiority: why ITT is not the safe analysis

Per-protocol
as actually treated

Intention-to-treat
20% did not take the drug



In a superiority trial

- Dilution makes the two arms look alike
- You fail to show a difference that is really there
- The error is on the safe side - conservative

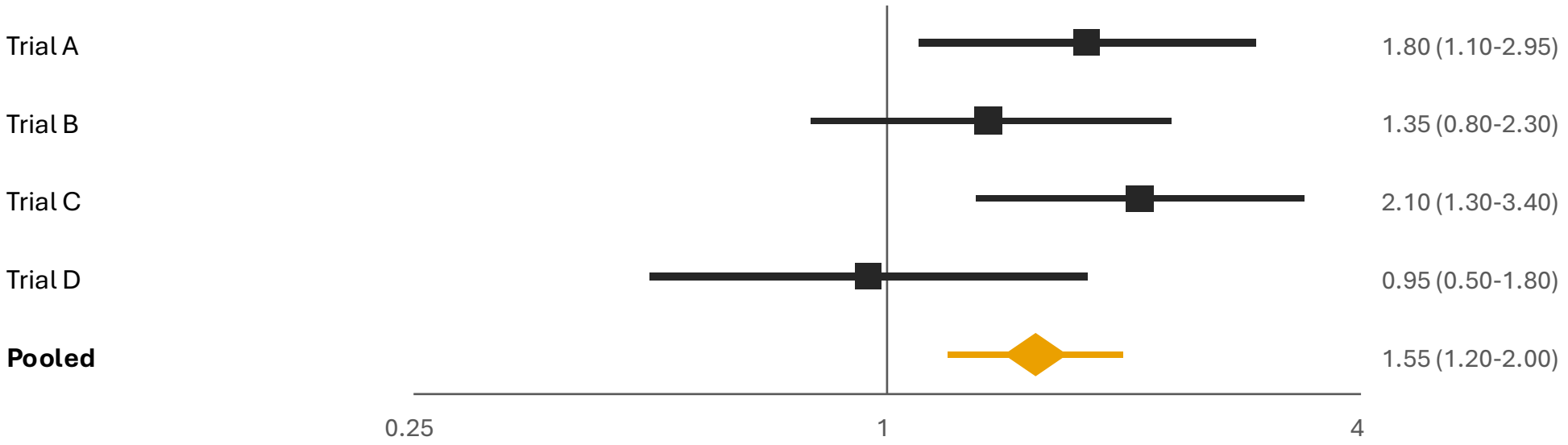
In a non-inferiority trial

- Dilution makes the two arms look alike
- But looking alike is exactly the conclusion you want
- The error is on the dangerous side - anti-conservative

The sloppier the trial, the easier non-inferiority becomes

What dilutes: non-adherence, crossovers between the arms, drop-outs, imprecise outcome measurement.
Per-protocol avoids the dilution but breaks randomisation - adherence is prognostic, so its bias has no predictable direction. Report both

Systematic review and meta-analysis



The unit of analysis is other studies - conceptually cross-sectional and observational

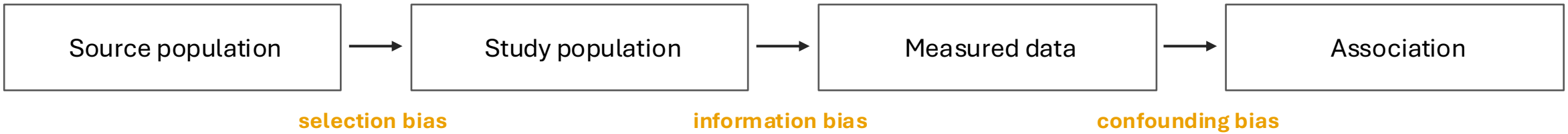
Pooling adds precision; it can never repair the validity of the included studies

There is no pyramid of evidence

Case report	Cross-sectional	Cohort	Case-control	Trial
the new and the rare	*prevalence, diagnosis*	*incidence, prognosis*	*rare outcomes, efficiently*	*therapy under equipoise*

A toolbox, not a hierarchy - no single instrument answers every question

Internal validity: three biases



Selection bias

- Who ended up in the study
- Enters at recruitment and at drop-out

Information bias

- How exposure and outcome were measured
- Random dilutes, differential distorts

Confounding bias

- What else explains the association
- A shared cause of exposure and outcome

Selection bias

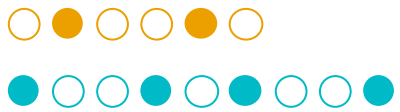
Source population



exposed

unexposed

Study population - who took part



developed the outcome did not

Non-participants

?

You never see them - and you cannot know how they would have compared

Selection bias is

- a difference between participants and non-participants,
- that differs across the exposed and the unexposed,
- and that is related to the outcome

Selection bias in dentistry

Where the patients come from	Who is missing	What it does to the result
Implant survival taken from your own recall list	patients who stopped attending	failures leave the practice - survival looks too good
Postal survey on smoking and tooth loss	smokers with many missing teeth rarely send the form back	the association is diluted - smoking looks less harmful
Case-control on peri-implantitis, controls taken from the hygiene recall list	the controls are healthier and smoke less than the source population	the relative risk comes out too large
Peri-implantitis measured in a survey of implants still in the mouth	implants that had already failed and been removed	the worst outcomes are gone - risk factors look weaker than they are

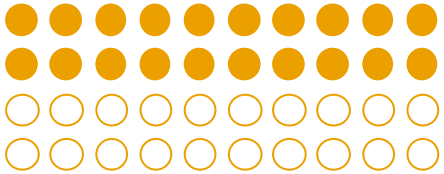
Selection bias needs both: participation differs between exposed and unexposed, and it is related to the outcome

Row 1 healthy-attender effect, row 2 non-response bias, row 3 control-selection bias, row 4 survivor bias.

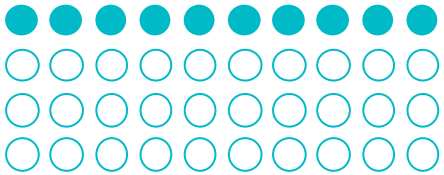
Selection bias: a numerical example

All implant patients - the truth

smokers



non-smokers



20/40 = 0.50 vs 10/40 = 0.25

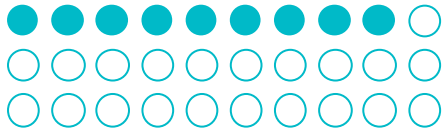
RR = 2.00 smoking doubles the risk

Only those who came back to recall

smokers



non-smokers



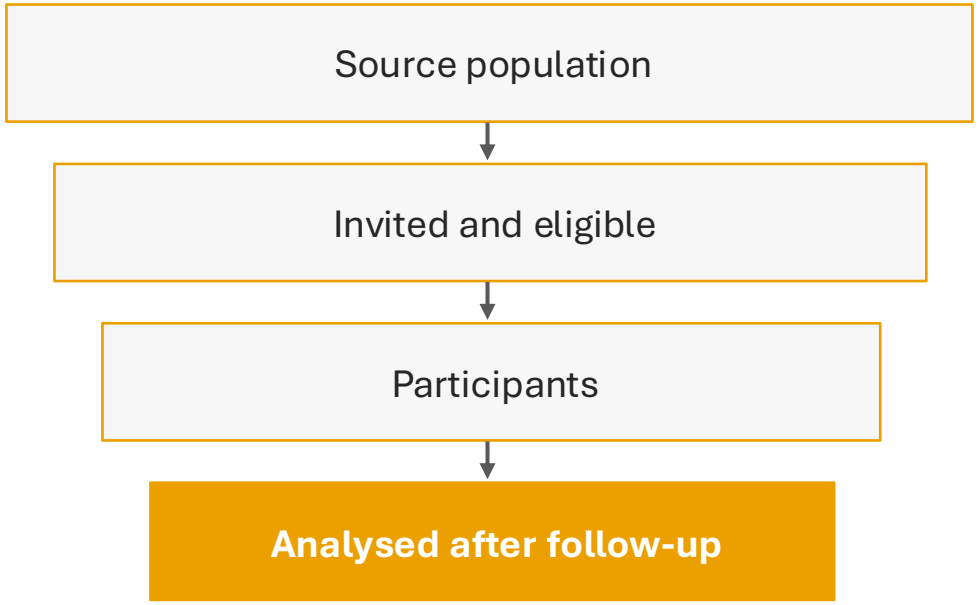
5/20 = 0.25 vs 9/30 = 0.30

RR = 0.83 smoking looks harmless

● implant lost ○ still in function

Three of every four smokers who had lost an implant stopped attending. A real doubling of the risk has turned into an apparent benefit of smoking.

Preventing selection bias



How to prevent it - in practice

- Define the source population before you recruit
  - every implant placed in 2019-2021, not everyone who came back
- Case-control: take the controls from the study base
  - controls from the same clinic and the same years as the cases
- Minimise loss to follow-up, and write down who was lost
  - short recall intervals, a reminder call, a note of every non-attender
- Compare those who dropped out with those who stayed
  - age, smoking, baseline bone level - and say so in the paper

Selection bias is a design decision, not an analysis decision

Information bias (misclassification)

Non-differential (random)

- Random measurement error in exposure, outcome or confounder
- Dilutes the effect towards the null
- Whatever you find is an underestimate

Differential (non-random)

- Error in the determinant is related to the outcome, or vice versa
- Asymmetry between the groups compared, in data collection
- Leads to under- or overestimation
- Direction cannot be predicted

Recall bias is the classic differential case: mothers of children with a birth defect remember dental radiographs, mothers of healthy children do not.

Misclassification in dentistry: random or differential?

Random (non-differential)

- Probing depth read to the nearest millimetre - the same error in smokers and non-smokers
- Brushing frequency from a questionnaire: everybody exaggerates, both groups equally
- Caries scored from bitewings by an examiner who does not know the group

Dilutes the association towards the null

whatever you find is an underestimate

Differential (non-random)

- The examiner knows the patient smokes and probes more carefully - diagnostic suspicion
- Patients whose implant failed remember their bruxism better - recall bias
- The implant group is recalled three times a year with radiographs, the control group once - you find more because you looked more

Over- or underestimates

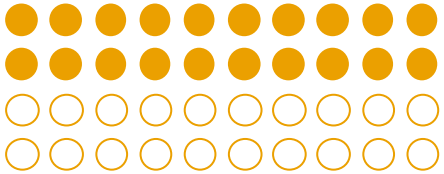
the direction cannot be predicted

Blinding and symmetrical data collection turn the right-hand column into the left-hand one

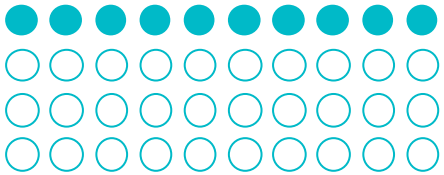
Information bias: a numerical example (non-differential)

Smoking recorded correctly

smokers



non-smokers

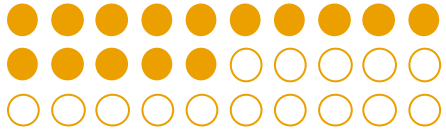


20/40 = 0.50 vs 10/40 = 0.25

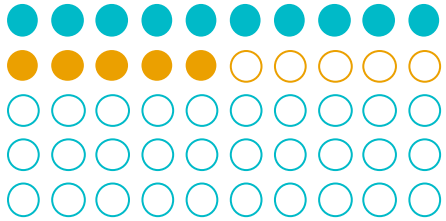
RR = 2.00

A quarter of smokers deny smoking

smokers



non-smokers



10 deniers: 5 with and 5 without implant loss – the same rate in both

15/30 = 0.50 vs 15/50 = 0.30

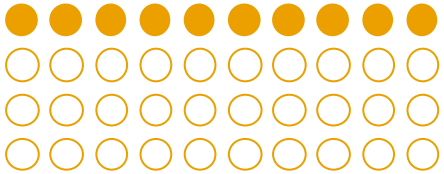
RR = 1.67

Filled dots: implant lost. The outcome plays no role in who denies – non-differential – yet RR falls from 2.00 to 1.67: random misclassification dilutes towards the null.

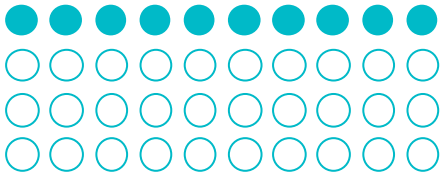
Differential misclassification: recall bias

What actually happened

radiograph in pregnancy



no radiograph

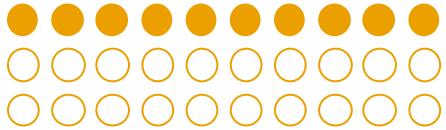


10/40 = 0.25 vs 10/40 = 0.25

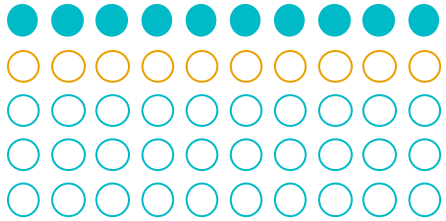
RR = 1.00 no association

What the interview records

reports a radiograph



reports none



10 mothers of healthy children forget their radiograph – mothers of affected children forget nothing

10/30 = 0.33 vs 10/50 = 0.20

RR = 1.67

Filled dots: child with a birth defect. Only mothers of healthy children forget – the error depends on the outcome – and a true null becomes RR = 1.67

Solutions: information bias

Non-differential error – make it small

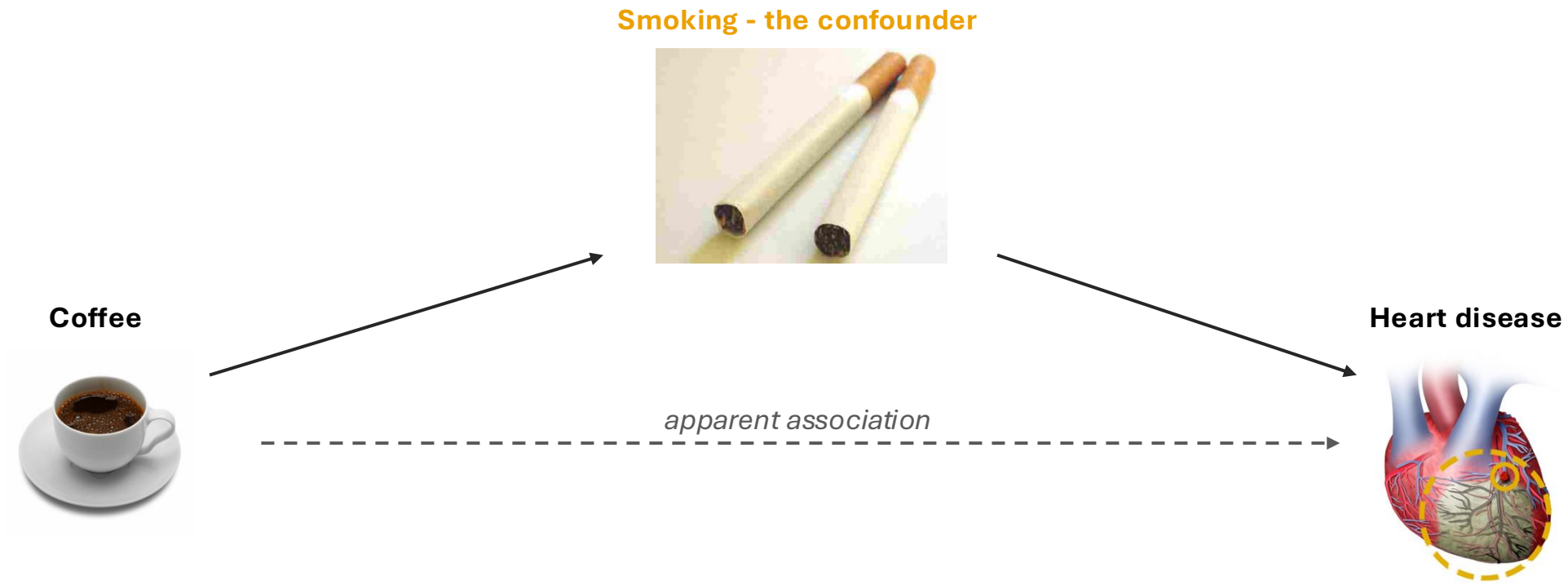
- Continuous scale (millimetres)
  - calibrate and standardise: same probe, same force, same angulation – for everyone
  - measure twice and use the average – and report examiner agreement
  - budget the rest in the sample size – only power is lost, no bias
- Categorical scale (yes/no) – dilution is a bias, no n repairs it
  - calibrated examiners – trained against a reference examiner until agreement is good, before the study
  - validated case definition – e.g. counting the bony walls of a peri-implantitis defect: write down beforehand where in-between cases go – one wall fully lost, the next only marginally: one-wall or two-wall?
  - adjudication – two readers score independently, a third settles disagreements
  - where the yes/no comes from a measurement, analyse the millimetres instead of the cut

Differential error – make it equal

- Blinding – single, double, triple
  - the patient; also the examiner; also the analyst
  - the radiograph reader must not know the treatment group
- Symmetrical data collection
  - same schedule, same instruments, same questions for both groups
  - equal recall intervals, equal radiographs, no probing one side harder
- Against recall bias: records instead of memory
  - take the radiograph exposure from the charts, not from interviews with the mothers
  - the chart was written before the outcome existed – it cannot remember selectively

Blinding and symmetry make the error random

Confounding: an everyday example



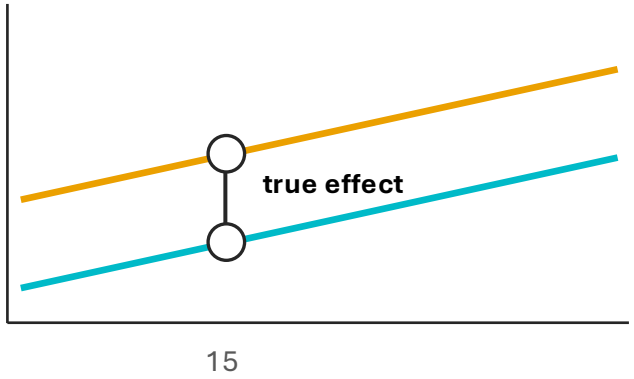
A confounder is a shared cause of the exposure and the outcome

Coffee drinkers smoke more - compare smokers with smokers and the coffee effect disappears

Confounding: compare at the same level of smoking

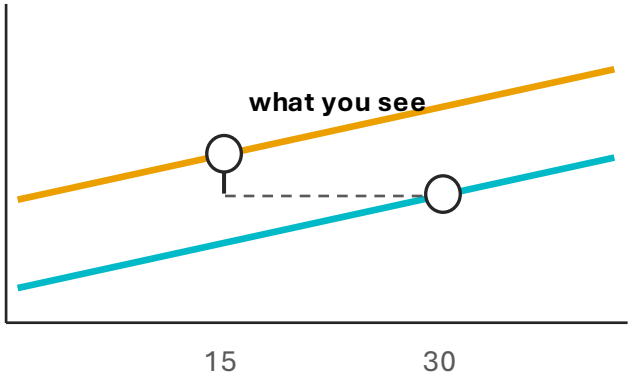
Diabetes No diabetes Diabetes, effect modified

No confounding



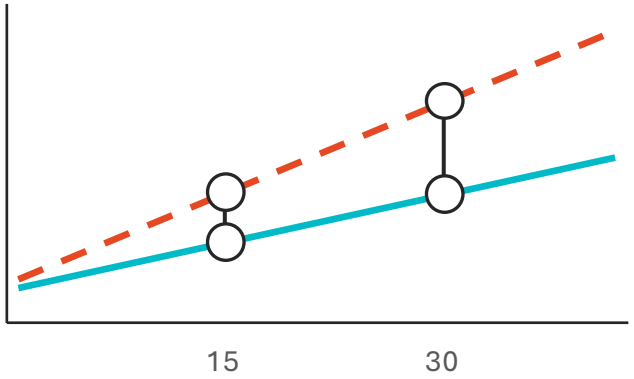
Both groups compared at 15 pack-years. The gap is the effect of diabetes and nothing else.

Confounding



Diabetics at 15, non-diabetics at 30 pack-years. The smoking effect is mixed in and the difference shrinks.

Effect modification



The lines are not parallel: the gap at 15 is not the gap at 30. No single adjusted number will do.

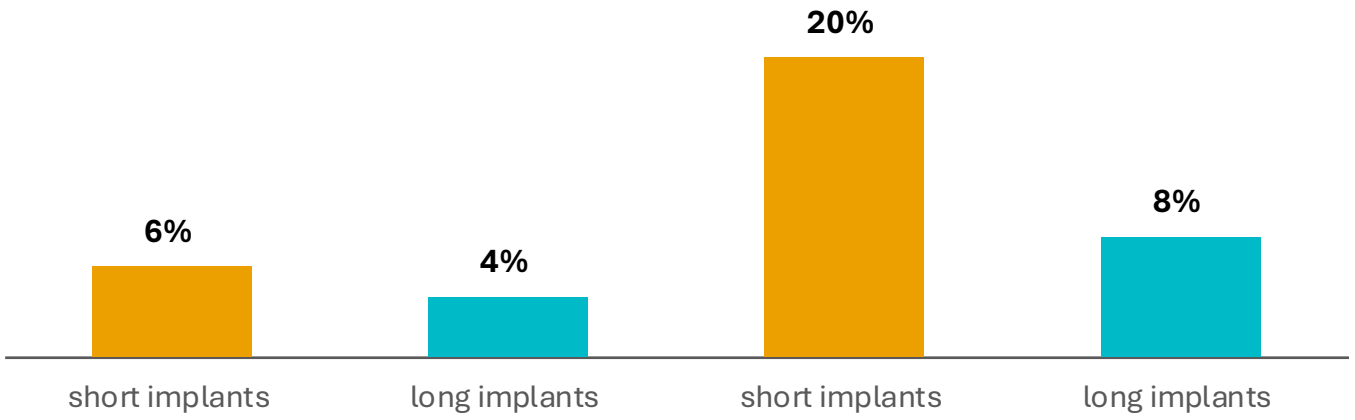
pack-years of smoking

Adjusting, stratifying and matching all force the comparison to the same level of smoking

Confounding: the crude estimate differs from the adjusted one. Effect modification: the strata differ from each other

Interaction and effect modification

Implant loss after 5 years (%), by bone turnover



No bisphosphonates

RR 1.5 - 2 points more

On bisphosphonates

RR 2.5 - 12 points more

Short implants are barely worse where bone turnover is normal - and clearly worse where remodelling is suppressed

The effect of implant length depends on the bone: one answer does not fit both groups

Report the strata separately - a pooled estimate would describe neither of them

Solutions for confounding

Design phase

- Restriction: study only non-smokers - simple, but you lose the question
- Matching: build comparability into the design
- Randomisation: the only method that also handles what you never measured

Analysis phase

- Stratification with Mantel-Haenszel pooling - the conceptual gold standard
- Multivariable regression: several confounders at once
- Change-in-estimate: keep the variable if the effect moves by more than 10%

Design deals with confounders known and unknown; analysis only with the ones you measured

In efficacy studies the classic case is confounding by indication: the sickest patients get the treatment. First the methodology, then the statistics - a regression model cannot tell you which variables belong in it.

Antepartum Dental Radiography and Infant Low Birth Weight

Philippe P. Hujoel, PhD
Anne-Marie Bollen, PhD
Carolyn J. Noonan, MS
Michael A. del Aguila, PhD

10/336 term low-birth weight with exposure > 0.4 mGy =3%

OR = 3.61 (95% CI: 1.46 – 8.92)

Context Both high- and low-dose radiation exposures in women have been associated with low-birth-weight offspring. It is unclear if radiation affects the hypothalamus-pituitary-thyroid axis and thereby indirectly birth weight, or if the radiation directly affects the reproductive organs.

Objective To investigate whether antepartum dental radiography is associated with low-birth-weight offspring.

Design A population-based case-control study.

Participants and Setting Enrollees of a dental insurance plan with live singleton births in Washington State between January 1993 and December 2000. Cases were 1117 women with low-birth-weight infants (<2500 g), of whom 336 were term low-birth-weight infants (1501-2499 g and gestation ≥37 weeks). Four control pregnancies resulting in normal-birth-weight infants (≥2500 g) were randomly selected for each case (n=4468).

Main Outcome Measures Odds of low birth weight and term low birth weight by dental radiographic dose during gestation.

Results An exposure higher than 0.4 milligray (mGy) during gestation occurred in 21 (1.9%) mothers of low-birth-weight infants and, when compared with women who had no known dental radiography, was associated with an adjusted odds ratio (OR) for a low-birth-weight infant of 2.27 (95% confidence interval [CI], 1.11-4.66, *P*=.03). Exposure higher than 0.4 mGy occurred in 10 (3%) term low-birth-weight pregnancies and was associated with an adjusted OR for a term low-birth-weight infant of 3.61 (95% CI, 1.46-8.92, *P*=.005).

Conclusion Dental radiography during pregnancy is associated with low birth weight, specifically with term low birth weight.

JAMA. 2004;291:1987-1993 www.jama.com

Where is the bias?

- **The paper**
  - “Antepartum Dental Radiography and Infant Low Birth Weight” – Hujoel et al., JAMA 2004;291:1987–93. Hypothesis: irradiating the maternal thyroid lowers birth weight
- **Design**
  - case-control in a dental insurer, Washington State 1993–2000: 1117 low-birth-weight births vs 4468 controls; exposure from the insurance records – procedure codes × published doses, no interviews
- **Result**
  - dose above 0.4 mGy: crude OR 1.80, adjusted 2.27 (95% CI 1.11–4.66) – based on only 21 exposed cases
- **Why it is not selection bias**
  - cases and controls come from the same insured births, enrolled before anyone knew who was exposed, and the record linkage is blind to radiography
- **What the records do rule out**
  - the exposure was written into the charts before the outcome existed – charts cannot recall selectively, so recall bias is gone
- **What they do not rule out**
  - the paper’s own audit found 93% agreement for radiographic procedures, and the doses are published averages, not measurements
  - with three dose groups cut from a dose nobody measured, error need not dilute: collapsed into “above 0.4 mGy versus none” it can turn differential and push the estimate up

A clean design does not end the critique – it decides where it moves next

Periodontal Infection as a Possible Risk Factor for Preterm Low Birth Weight

Steven Offenbacher, Vern Katz, Gregory Fertik, John Collins, Dorvck Bovd. Gavle Mavnor. Rosemary McKaig, James Beck

Abstract

Periodontal diseases are Gram-negative anaerobic infections that can occur in women of childbearing age (18 to 34 years). In the present investigation we sought to determine whether the prevalence of maternal periodontal infection could be associated with preterm low birth weight (PLBW), controlling for known risk factors and potential covariates. A case-control study of 124 pregnant or postpartum mothers was performed. PLBW cases were defined as a mother with a birth of less than 2,500 g and one or more of the following: gestational age <37 weeks, preterm labor (PTL), or premature rupture of membranes (PROM). Controls were normal birth weight infants (NBW). Assessments included a broad range of known obstetric risk factors, such as tobacco use, drug use, alcohol consumption, level of prenatal care, parity, genitourinary infections, and nutrition. Each subject received a periodontal examination to determine clinical attachment level. PLBW cases and primiparous PLBW cases (n = 93) had significantly worse periodontal disease than the respective NBW controls. Multivariate logistic regression models, controlling for other risk factors and covariates, demonstrated that periodontal disease is a statistically significant risk factor for PLBW with adjusted odds ratios of 7.9 and 7.5 for all PLBW cases and primiparous PLBW cases, respectively. These data indicate that periodontal diseases represent a previously unrecognized and clinically significant risk factor for preterm low birth weight as a consequence of either PTL or preterm PROM. *J Periodontol 1996;67:1103–1113*.

Confounding by indication – in the JAMA study

- **The disease chooses the exposure**
  - nobody radiographs a healthy mouth: caries, abscesses and periodontitis send the mother to the X-ray – and periodontitis was itself linked to low birth weight (Offenbacher 1996: adjusted OR 7.9, three times the radiation finding)
- **The paper carries its own fingerprint**
  - Mothers of low-birth-weight infants had fewer preventive procedures (87 vs 103 per 100 woman-years), and term-LBW mothers more endodontic ones (5.5 vs 3.0) – the profile of sicker mouths
- **Adjustment could not close it**
  - fourteen covariates from the birth certificate, none about the mouth; adjusting for procedure counts (OR 2.54) counts visits, not disease severity
- **Structural pathway**
  - Dental disease → radiograph and dental disease → low birth weight; the study measured no oral-health variable, and later studies did not confirm the radiation link, i.e. the study did not adjust for oral health

Structural confounding survives any sample size

Confounder misclassification (smoking) – in the JAMA study

The rule – error in the exposure

Random misclassification of the exposure dilutes the association towards the null

Whatever you find is an underestimate (slide "Information bias")

The exception – error in a confounder

Random misclassification of a confounder leaves residual confounding

You adjust for "smoking" but only remove part of the smoking effect – the estimate stays pushed away from the null

In the JAMA study

1 Smoking is a confounder

Smokers have sicker mouths → more radiographs; smokers have more low-birth-weight babies

2 Smoking was measured badly

Taken from the birth certificate – about 1 in 4 smokers is recorded as a non-smoker

3 The hidden smokers are not spread evenly

They sit mostly in the radiograph group – so adjusting for recorded smoking leaves part of the smoking effect inside the radiation OR

Even if smokers lie equally often in both groups, the lie is concentrated where the smokers are – in the radiograph group.

Adjusting for a badly measured confounder is only half an adjustment

Small numbers – in the JAMA study

As published

	> 0.4 mGy	No radiograph	Total
Term low birth weight	**10**	287	**297**
Controls	**46**	4003	**4049**

Odds ratio = (10 × 4003) / (287 × 46) = 3.0

The whole association rests on 10 exposed cases

If only 3 of the 10 were recorded wrongly

	> 0.4 mGy	No radiograph	Total
Term low birth weight	**7**	290	**297**
Controls	**46**	4003	**4049**

Odds ratio = (7 × 4003) / (290 × 46) = 2.1

95 % CI 0.94 to 4.69 – no longer significant

3 of these 4346 women (297 + 4049) decide whether the paper has a finding.

Instability – the number is shaky

10 women on one end, 4000 on the other

Moving one of the 4000 changes nothing; moving one of the 10 swings the result. The wide interval (1.46 to 8.92) is the statistics admitting this.

Inflation – the number is pushed up

Sparse cells bias the odds ratio away from 1 – and every added covariate pushes it further

The paper shows it: 3.05 crude → 3.61 with 14 covariates. Below ~20 per cell, read the odds ratio as a direction, not a size.

A finding that ten women carry is a hypothesis, not a risk estimate

How a clinical trial handles the three biases

- Selection bias
  - not present at study entry – randomisation gives every patient the same route in
  - only possible through selective loss to follow-up – which is why trials chase every patient
- Information bias
  - blinding – patient, examiner, analyst
- Confounding bias
  - randomisation – removes it with a sufficiently large sample size, measured and unmeasured alike
  - multivariable modelling – mainly in small trials or subgroup analyses

Randomisation and blinding are built in – selective loss to follow-up is still possible

Quiz (true or false)

In a descriptive study, random sampling from the source population is preferred
In a case-control study, information bias is non-random if the assessment of the exposure differs between cases and controls
The attributable risk is the same as the relative risk difference
Removing measurement error in the exposure primarily improves the precision of the study
Effect modification occurs when the effects across two strata differ from each other

Quiz (continued)

A confounder is a shared cause of exposure and outcome

An RCT can per definition not suffer from confounding, because exposure is randomly allocated

Solutions for confounding in observational studies include matching, restriction and multivariable analysis

If the findings of a trial in men are not generalizable to women, there may be effect modification

If extremely sick individuals refuse to enrol in an RCT, it is a matter of external validity

Take-home messages

Write the occurrence relation first: f(determinant) = outcome

The design is the toolbox - there is no pyramid of evidence

Three biases: selection, information, confounding - all three are design problems

Statistics cannot repair a study; it can only describe what the design produced